APRIL 6TH, 2023

FINANCE OPERATIONS RESEARCH EVENT

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CHAPTER

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I. EXECUTIVE SUMMARY

ABOUT THE BUSINESS

Based in Washington state, Sound Credit Union is a financial institution that offers a range of banking services, including checking/savings accounts, loans, and mortgages. SCU is committed to serving the local community and offers a variety of financial education resources. They have multiple branch locations in Western WA and offer online and mobile banking services for the convenience of its members.

sound credit union
Invested in We™

Members. Employees. Community.

Standing with you through all waves of life.

TARGET MARKET

PRIMARY
- Age: 27 - 58
- Middle-income
- Financially-active
- Urban locations in WA

SECONDARY
- Age: 16 - 26
- Students
- Suburban locations in WA

RESEARCH

PURPOSES

- Determine methods to enhance the business's current data collection and analysis
- Identify customer concerns over existing data collection and utilization practices and general privacy concerns
- Understand the benefits to the business and customer of the implementation of hyper-personalization using artificial intelligence

METHODS USED

Interviews

Interviewed Senior VPs of Sound Credit Union

Surveys

Surveyed Members and Potential Members

Online Research

Researched Costs of Hyper-personalization

FINDINGS & CONCLUSIONS

Findings	Conclusions
SCU's data collection is unintegrated in various pipelines of operation	Sound could expand data collection, and make pipelines to be realtime and integrated
Customers want to understand how their data is used by banks, and have elevated concerns on privacy	Sound can better educate users about when they use their personal data, and how it is being used
SCU can do more to include under-tapped demographics of the community	Sound can develop personalized services to better serve underserved communities

OBJECTIVES OF STRATEGIC PLAN

1) Reduce Privacy Concerns

2) Improve Data Collection Methods

3) Build inclusion-first principles in AI/ML solution to reach under-served communities.

4) Build a scaleable & cohesive hyper-personalization solution for SCU

PROPOSED STRATEGIC PLAN

In order to improve Sound Credit Union's business data collection strategies and practices, we propose the "BOND Together" plan with the intent to grow Sound Credit Union by implementing hyper-personalization services for a stronger bond with each customer. Our proposed strategic plan is outlined by the acronym BOND.

	Build	**Optimize**	**Network**	**Drive**
	Trust With Our Members	Data collection & quality	With Underserved Communities	For scale With hyper-personalization
ACTIVITIES	Public Board Meetings	Optimize Data Models	Help Students	Deliver Hyper-Personalization
	Closed Member Feedback Loop	Training Existing Staff	Support Service Workers	Streamline Operational Processes
	Community Events	Rollout of revamped personalization Model	Target Low-Income Demographics	Redesign Online Platforms
TIMELINE	2023 Q1 Starting Jan - Mar	2023 Q2 Starting Apr - May	2023 Q1-2 Starting Feb - Jul	2023 Q1-2 Starting Mar - Apr
KPIs	Customer Participation %	Customer Engagement Rate, Click-Through Rate	Net new signups / demographic	Reviews/ Engagements collected
BUDGET	$21,200	$124,200	$18,000	$100,000

CONCLUSION

TOTAL COST: $263,400

The purpose of our project is to provide Sound Credit Union with better ways to collect and analyze the data they receive about their customers through transparent hyper-personalization technologies, ultimately to enhance and improve the company's services. We plan for an initial budget of $263,400 to implement our proposed strategic plan and the "BOND together" initiative, and project an ROI of 90% from the investment, given the opportunities available with the implementation of hyper-personalization.

II. INTRODUCTION

A. Description of the business or organization

Started by 51 telephone workers in the 1940s, Sound Credit Union is a United States credit union headquartered in Seattle, Washington and serves the people of the Northwest. Sound Credit Union is one of Washington State's largest credit unions, driven by the purpose of standing with members, employees, and the community through all waves of life. Unique in that all 29 branches are full-service, Sound provides nearly 160,000 members with authentic and trusted financial support and services around the Puget Sound.

Members. Employees. Community.

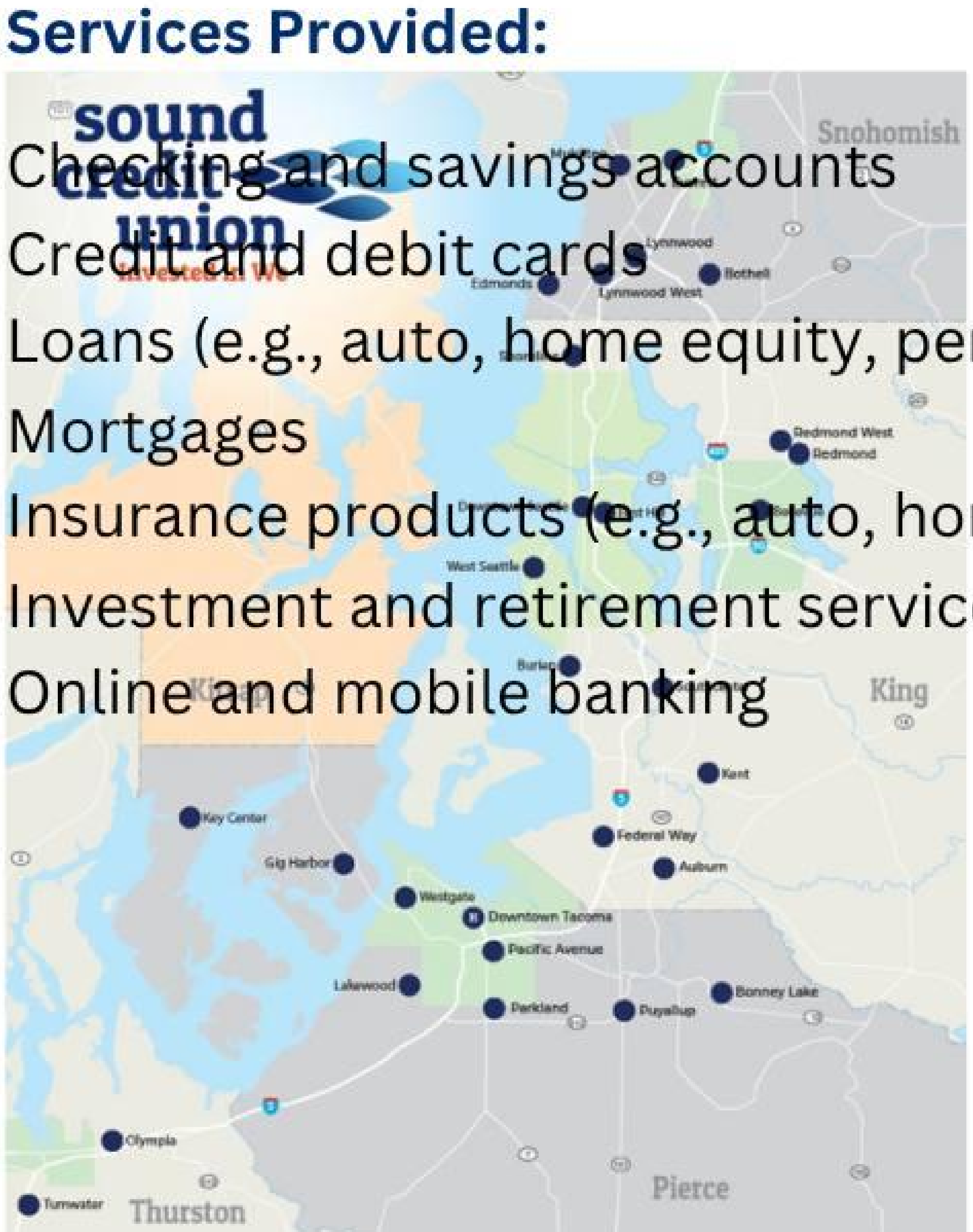
Standing with you through all waves of life.

Unlike large public banks, Sound is built as a **not-for-profit**, member-owned financial institution designed to help the community around the Washington Puget Sound region. Sound is different from publicly listed banks in that they do not need to cater to shareholder dividends by charging higher interest rates. Credit union banking instead entail their customers as equity shareholders, and provide members ownership stake in operational affairs.

Sound Credit Union has a rich history, starting out with $200 and 5-10 people coming together and pooling money to loan money to people, growing into managing $2.5 billion in assets under management. However, Sound still exists to provide financial services back to members of the cooperation and all members essentially pool funds together to lend money to each other at a massive scale. Whatever revenue is made is given back to their members in the forms of lower interest rates, lower than the bank competitors. Because of this, Sound always reaches out to the community to better understand where they can grow, preferring to refer their customers as "members". Richard Romero is the current serving CEO of Sound, and he employs 372 staff.

Services Provided:

- Checking and savings accounts
- Credit and debit cards
- Loans (e.g., auto, home equity, personal, business)
- Mortgages
- Insurance products (e.g., auto, home, life, health)
- Investment and retirement services
- Online and mobile banking



$20.1M Direct Member Benefits

$2.5B AUM **160k** Members

320+ Local Organizations Supported

B. Description of the target market (demographics and psychographics)

PRIMARY **SECONDARY**

Demographics

Students
27-58 year-olds
16-26 year-olds
Economically Middle-Income
Digital Exclusive Customers

Psychographics

Supportive of the CommunityCurrent SCU Customers
Financially ActiveDissatisfied Traditional Bank Users
Wanting to be Financially Active

Geographics

Washington's Urban Areas:Washington's Non-Urban Areas:
KirklandLakebay
BellevueGig Harbor
Seattle Puyallup

Behavior

Loyal to local businessesWants to minimize banking
Appreciates lower feesexpenses
Maintains a simple financialLooking for introductory
portfolio finances

Customer Acquisition

By means of digital marketing, mailing lists	By means of digital marketing, community outreach events

All target market information was provided from our interview with Jennifer Reed, VP of Community and Public Relations at SCU.

C. Overview of the business or organization's data collection practices for business decision making

Like other financial institutions, SCU collects customer data for various purposes, including identity verification, lending adjudication, customer profile retention for regulatory purposes, and other minimal information on the customer and their intent to banking in compliance with jurisdictional and federally regulated laws. SCU puts a premium in ensuring customer privacy and safety in the use of this information following systemic controls, and processes such as data retention, purpose-based-policies, and archival. SCU shares this data with their BI (business intelligence) group with a 3rd party consortium, Facteus Inc., who shares member insights with SCU. Sound does not currently have a team of data analysts.

The data collected by SCU can be observed through the following taxonomies:

a) **Decision making**: Sound Credit Union collects data regarding *daily transaction information,* the *types of bills being paid inside and outside of the credit union*, their *credit history*, and other personal information of this sort to see if the member is credit worthy by determining their FICO score. Lenders may use credit report information to decide whether you can get a loan and the terms you get for a loan (i.e. the interest rate charged). Sound tracks the majority of information when members apply for their accounts, rather than looking into their data when the account is opened. Sound may look at data from members who have already opened their accounts in order to track how many deposits are made on average on a macro-level, who the most frequent payees are, and general transfers. Sound primarily collects this data through customer interactions and services.

b) **Marketing**: Sound collects customer data in *general demographics* and media content-tracking to gain insights with CRM (Customer Relationship Management) technologies, services, and other internal tools, and to better tailor content for their marketing material and advise for tailored products and services for their customers. For example, seeing as younger people more often set up accounts digitally, SCU can use this information to personalize younger users towards digital adoption. CRM systems start by collecting customer's information by requesting *email, telephone #,* and *social media data,* across multiple sources and channels, and organizes this collected data and manages day-to-day customer activities and interactions. Sound tracks content with social media platforms like Facebook, Instagram, and Twitter for advertising and understanding efficacy of their advertising content. Sound also reaches out to customers through *email*, and personalizes them with each member's *first and last name*.

c) **User Security**: Although SCU collects data for a variety of reasons, none are as important to them as the information and metrics tracked to ensure customer safety and privacy. SCU tracks many data points to verify customer identification, such as their *driver's license information* and *SSN,* and also tracks *information regarding their money transferring*, *account logins, phone types*, *operation systems,* and *IP addresses of the device used* to make sure there is no fraudulent account activity. Sound primarily collects this data from their website.

Regulatory Compliance:

Sound Credit Union's data collection always strictly follows the Bank Secrecy Act (BSA) and the Office of Foreign Assets Control (OFAC) regulations. Sound Credit Union does not share private information with third parties, but may use the information to benefit themselves to market to individuals. Sound Credit Union also offers Federal Deposit Insurance Company (FDIC)-like protection through the National Credit Union Share Insurance Fund (NCUSIF).

BSA: "Requires U.S. Financial Institutions to assist U.S. government agencies to detect and prevent money laundering" - Financial Crimes Enforcement Network (US Government Website)

OFAC: "Administers and enforces economic sanctions programs primarily against countries and groups of individuals, such as terrorists and narcotics traffickers" - US Department of the Treasury

III. RESEARCH METHODS USED IN THE STUDY

A. Description and rationale of research methodologies selected to conduct the research study

Purposes of Research

1. Determine methods to enhance the business's current data collection and analysis
2.Identify customer concerns over existing data collection and utilization practices and general privacy concerns
3. Understand the benefits to the business and customer of the implementation of hyper-personalization using artificial intelligence

PRIMARY RESEARCH

Research Method	Description	Rationale
Interviews	a) Interview with Jennifer Reed, VP of Community and Public Relations b) Interview with Perry Maxfield, VP of Digital and Innovation Delivery c) Interviewing another credit union that currently use hyper-personalization	We chose the interview research method to receive data from experts by asking questions in real-time. The purpose of interviewing two VPs was to determine Sound's existing data collection practices from experts who actively make decisions in fields of a) Marketing and b) Financial Transactions and Opening Accounts, in order to enhance current practices, as well as better understand SCU's business model, target market, etc. The purpose of interviewing the other credit union (c) was for market intelligence research, comparing their data collection with that of SCU and to determine potential benefits to the business of the implementation of hyper-personalization using AI.
Surveys	a) Survey to Sound Credit Union customers regarding potential concerns with Sound's data collection practices b) Survey to adults aged 27 - 58 in the Puget Sound region who are not current Sound customers, regarding their potential concerns with Sound's data collection practices.	We chose the survey format to receive input from a large population and for easy data analysis from digital tools. The purpose of sending surveys to various Sound customers (a) was to determine customer concerns of Sound's data collection and utilization practices. We also sent a similar survey to other adults that fit Sound's target market (b) to understand their opinions as potential customers about the data collection practices that Sound Credit Union uses.

SECONDARY RESEARCH

Research Method	Description	Rationale
Online Research	a) Online research of Sound Credit Union's Privacy Policy by analyzing Sound's website b) Online research in the costs of implementing AI and hyper-personalization by reviewing data and reports c) Online research on general customer concerns about hyper-personalization approaches in the financial industry	We chose the online research format to research vast amounts of information with different perspectives, eliminating bias in our analysis. The purpose of conducting online research was to research the precise policies and practices Sound Credit Union uses to collect data (a), and to determine the benefits of using hyper-personalization in AI to both the business and the customer. We extended this online research to also study the costs of implementing AI and hyper-personalization (b), and other customer concerns (c) by reviewing data and customer reviews from financial businesses which currently use hyper-personalization services, to further determine the benefit to the business of implementing artificial intelligence hyper-personalization.

B. Process used to conduct the selected research methods

We conducted our secondary research before our primary research to learn more about Sound Credit Union and its business model. We wanted to first get a general understanding of the concepts of hyper-personalization and credit unions before researching specific topics and private information, which we were able to receive afterwards once we started conducting our primary research.

Secondary Research - offline research

Primary Research - direct interactions with SCU

(Nov 2022)

(Dec 2022)

Understanding the background
Building understanding of credit unions, and their clientele, general banking practices on privacy and compliance, bank's public facing commitments on data privacy

Researched industry trends
Review current market trends in data personalization - Deloitte, Gartner research on where the industry is headed in personalized banking

Interview #1: Jennifer Reed, VP of Community and Public Relations
Learning what's unique about SCU and its customers, current customer outreach programs, personalization programs at SCU

Interview #2: Perry Maxfield, VP of Digital and Innovation Delivery
Role of machine learning, data collection pipelines and policies, SCU market demographics, high level understanding of customer segmentation

Technology Research
Researching AI and ML tools, understanding application of AI tools, building prototypes on ML to evaluate performance of models, and data quality and validation, marketing trends

Interview #3: Marangal Domingo, CFO of a leading credit union based at Seattle
Banking regulations, hyper-personalization trends at industry, competitive landscape, customer viewpoints towards personalization

Understanding personal banking regulations
Researching customer intent on banking through news, recent concerns and litigations on privacy, social responsibility, CCPA, BSA, and OFAC

Customer Surveys
We sent a Google Forms survey to ~100 banking users to learn on their sentiments with regards to banking, personalization and privacy. We received a high number of responses indicating user propensity and perception to data collection

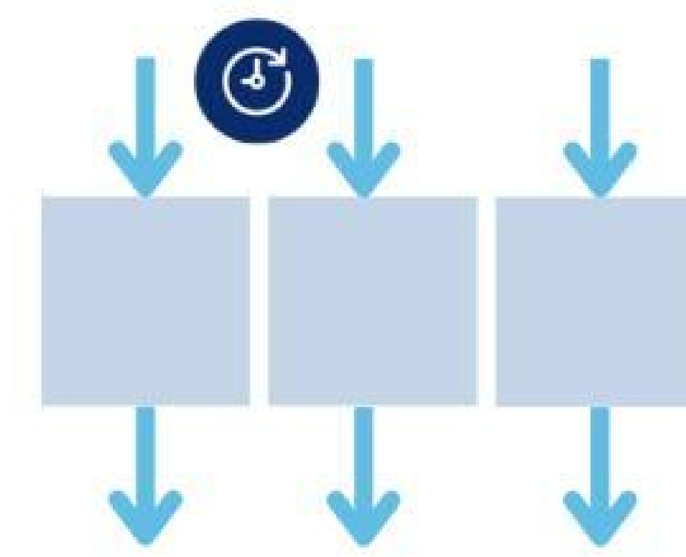
IV. FINDINGS AND CONCLUSIONS OF THE STUDY

A. Findings of the Research Study

FINDING 1: SCU collects limited amounts of data that is not integrated or collected in real-time

Primary Research (Interviews)

1. From our interviews with the VPs of Public and Community Relations, and Digital Innovation and Delivery, we found out that SCU currently uses personalization in their marketing with targeted advertising with internal CRM tools. Sound currently tracks this personal data to address messages to the recipient and tailor content, but this data analysis is not currently in real time. The lack of realtime data integration causes SCU to lose out on time-sensitive presentment of opportunities.



Each line of business collects and processes data independently

2. From our interview with a representative of competitor credit union bank who had already invested in personalization efforts, we found that they were benefiting from their implementation of hyper-personalization in real-time analysis for tailoring services to customers. He told us that "with this new technology, businesses can gain costs savings up to 30% and revenue increases up to 20%. Additionally 91% of customers say they are more likely to pay for brands that offer recommendations relevant to them".

Potential 20%+ lift in revenues

Potential 30%+ cost savings in operational efficiencies

Potential 90%+ increase in customer engagement on personalization

Secondary Research (Market study)

Deloitte's CMO surveys indicate high potential for companies that implement personalization in the right way. The greater availability of data, and customer's favorability to work with companies that understand them better provide a highly positive outcomes where hyperpersonalization becomes the differentiator, and sometimes the tool to survive the needs of the 21st century banking.

80% *of customers are more likely to purchase from a company that offers personalized experiences*

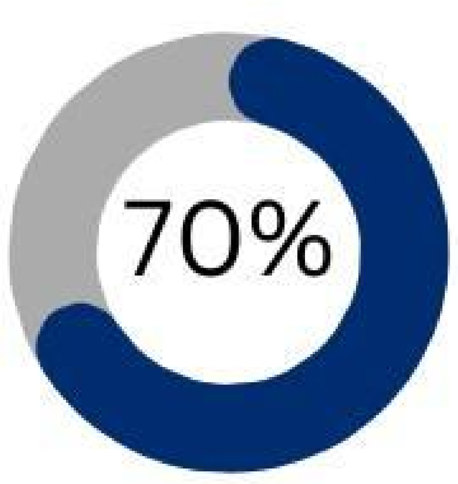
8x *return on investment on marketing spend*

FINDING 2: The majority of people surveyed raise ethical concerns with personalization practices and want to better understand what kinds of data they are sharing and how they are being used

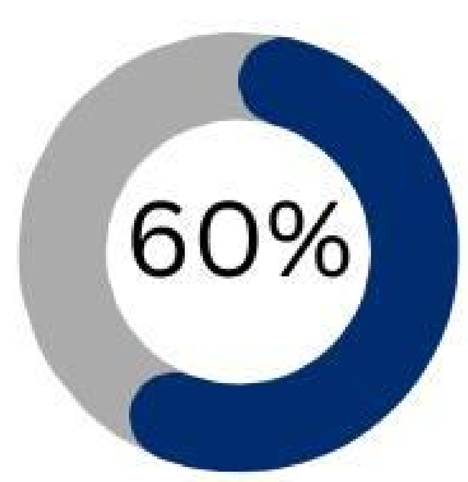
Primary Research- Surveys

From our surveys sent out to both existing and potential Sound customers, we found:

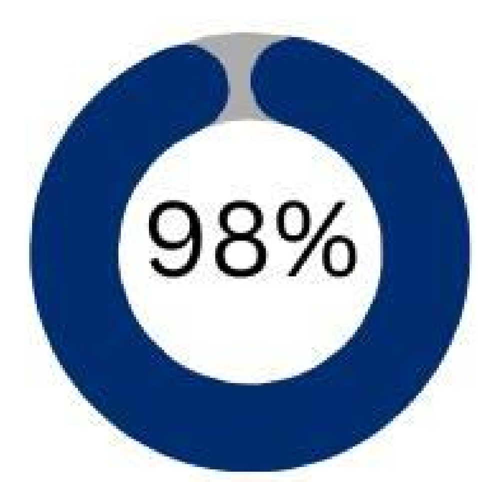
- 70% of people said that appreciate personalized recommendations and feel valued when relevant offers are sent to them
- 60% expressed discomfort with over-personalization
- People generally do not mind sharing personal information as long businesses do not use any other information that is not appropriate for them to have access to.
- 98% of all respondents said that they wanted to better understand the exact kinds of data being tracked from them and how it is being used, through transparent credit union communication.



appreciate relevant recommendations on offers



concerned on over-personalization encroaching privacy



of customers want to understand how their data is used by banks

FINDING 3: There is an untapped market that SCU could attract, specifically low-income demographics - disconnected with their mission to benefit the community

Primary Research (Interviews)

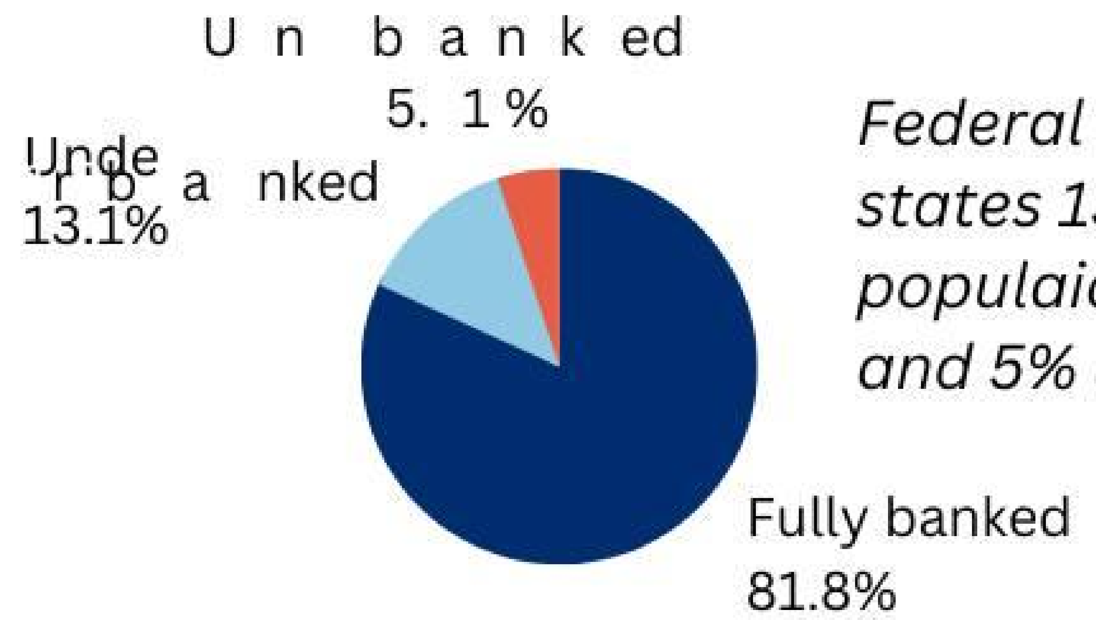
One surprising finding from our interviews regarding Sound Credit Union's target market breakdowns was that even though Sound prides their low interest rates and humanitarian appeal of working to better the Puget Sound community, they do not market specifically towards lower-income individuals, instead choosing to target middle-higher class full nesters and students. We believe this is a missed opportunity for Sound, because lower-income people would appreciate the lower interest rates and community spirit of them.

Secondary Research (market study)

From online research, we found targeting low-income households can provide the following benefits:

- Increased business opportunity
  - Can expand their customer base by providing solutions to the underserved
- Socially responsibility
  - Bringing in under served population who are traditionally alienated from banking aligns with the bank's corporate social responsibility policy and enhance its reputation
- Compliance with regulations
  - Regulators require banks to serve the credit needs of all communities, including low- and moderate-income neighborhoods, as part of their Community Reinvestment Act (CRA) obligations.

40% of WA residents have a household income < $70K



Federal reserve study states 13% of US populaion underbanked, and 5% unbanked

B. Conclusions based on the Findings

CONCLUSION 1: Sound Credit Union and their customers will greatly benefit from hyper-personalized services

Determined from our online research, interviews with other credit unions, and surveys, we found that a business's understanding of an individual's personal preference is beneficial to all; both the customer and business benefit from hyper-personalization. This is described below:

For Sound Credit Union	For Sound's Customers
Increased customer satisfaction	Improved relevancy of recommendations
Increased customer loyalty	More personalized customer service
Increased revenue	Increased convenience
Better services for customer needs	Improved user experience
Increased efficiency	Feeling of being valued as an individual

CONCLUSION 2: Sound can better educate users about when they use their personal data, and how it is being used

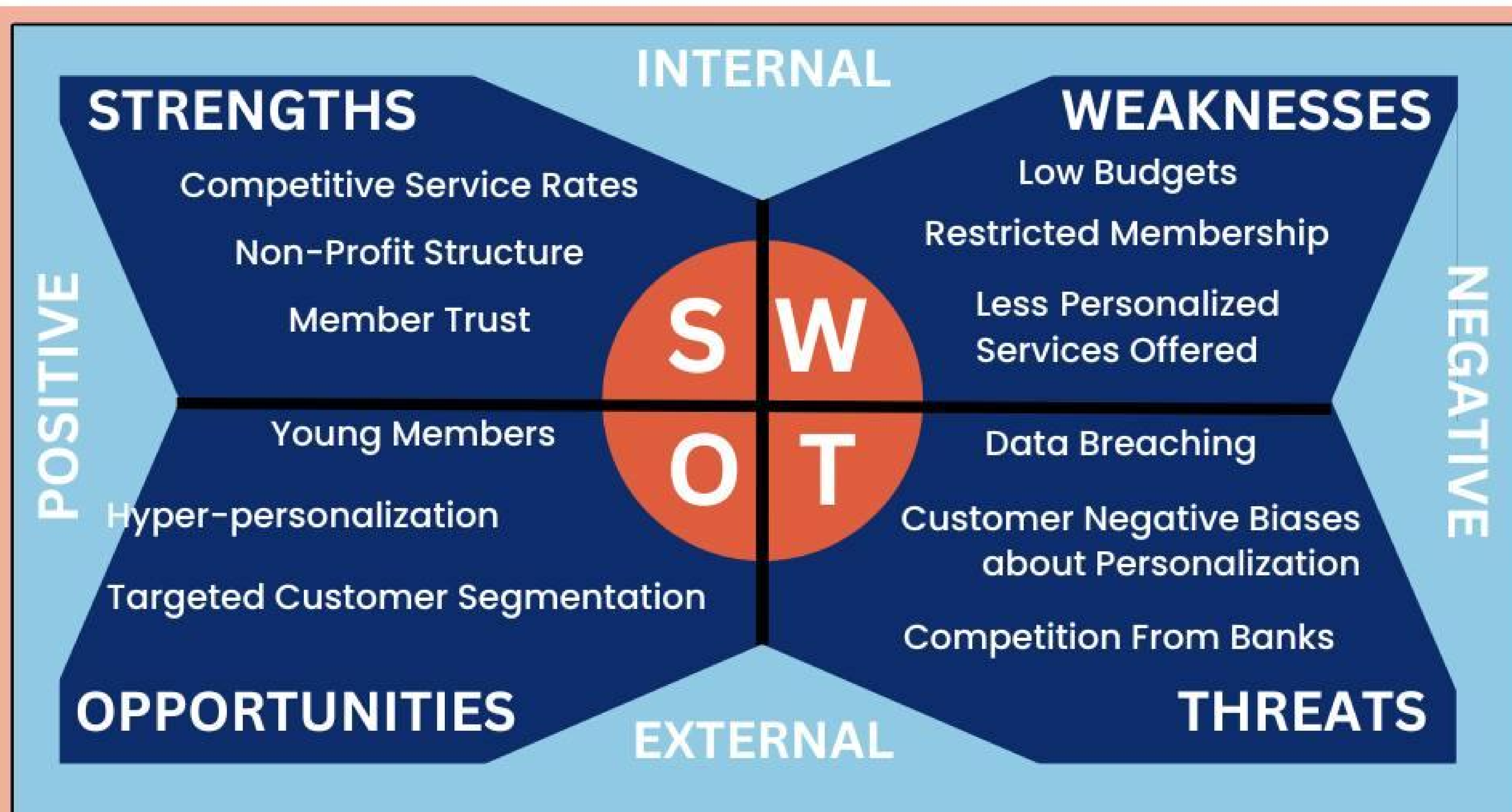
Most survey responders mentioned that they would appreciate better transparency between the business and the customers. Sound can differentiate from their competitors by being upfront about what data is collected, and the intent for doing so. With this, customers would be more likely to trust Sound with ethical data collection practices, and can demonstrate how Sound first serves the community. Additionally, Sound can share transparency reports with customers regarding data privacy and protection compliance.

CONCLUSION 3: Sound can develop personalized services to better serve underserved low-income families

Sound does not include lower-income families primarily in their target market, and are therefore missing out on a large, underserved market. Sound's mission to benefit the community seems to well support lower income, underserved individuals. Sound Credit Union can improve their customer segmentation data analysis tools, and can use these services to include specific demographics and psychographics, like lower income people and racial minorities. By focusing on similar underserved communities, Sound Credit Union can increase its customer base, improve customer satisfaction and loyalty, and also contribute to building financial inclusion.

SOUND CREDIT UNION COMPANY SWOT ANALYSIS:

Our SWOT analysis of SCU allows us to understand the competitive advantages and disadvantages we would have to take into consideration. This will be a key input into the strategic plan.



V. PROPOSED STRATEGIC PLAN

A. Objectives and rationale of proposed strategic plan

Our team's primary objective is to aid Sound Credit Union in the implementation of hyper-personalization when interacting with its members. Through the improvement of Sound Credit Union's practices around data collection and its uses, we hope to achieve and maintain a service ROI > 90%. Our strategic plan was designed to address the 4 objectives designated below.

OBJECTIVES	RATIONALE
Reduce privacy concerns	Our initial goal is to alleviate customer privacy concerns. Our research reveals that more than 60% of individuals are uneasy about excessive personalization, and almost all survey participants (98%) desire greater transparency from their banks regarding data collection practices. We will implement hyper-personalization with privacy-first principles, and providing Customers with transparency messaging on how their data is used anywhere user data collection is involved.
Expand data collection methods	Sound's limited data collection prevents them from fully implementing personalization, causing doubts among customers about the customized services and products they receive. Only 1/3 of survey respondents believe they receive personalized treatment. Research indicates that effective data collection and personalization are crucial in building strong customer relationships. Sound can enhance customer connections and generate substantial profits by enhancing and expanding their data collection methods.
Build inclusion-first principles in hyper-personalization	As we build our personalization modules on AI/ML, we will proactively guard against bias in applying decisions. Towards this end, we will expand our data pipelines to accommodate inclusivity, and train the models to be diversity-aware. This will directly contribute on solutions to better reach underserved segments and young users.
Build a scaleable & cohesive hyper-personalization solution for SCU	The writing is clear for banks of the 21st century. Customers expect their banks to understand their unique needs, and tailor solutions that work for them. We will apply Hyper-personalization through artificial intelligence (AI). in the areas of customer segmentation and targeting, and drive and maintain repeat engagement with customers and build loyalty over time. Hyper-personalization technologies will also enable the bank to competitive with the market trends adopted by its competitors. We will revamp the customer storefronts to be cohesive and scaleable.

B. Proposed activities and timelines

The "BOND together" initiative is a multi-step plan, outlining the actions necessary to achieve the objectives above and ultimately our goal of hyper-personalization in a timely manner, which we intend to be implemented by the upcoming fiscal year.

Build

Trust With Sound Members

Optimize

Data Collection

Network

With Underserved Communities

Drive

For Scale With Hyper-Personalization

Build

Trust With Our Members

Objectives: Reduce privacy concerns

In this phase of the project, we focus on activities towards building trust with members, and stakeholders. We will implement the confidence building measures to obtain alignment and approval from SCU members.

When activities will start

Public Board Meetings

MAR 2023

Sound Credit Union has 5 primary board meetings throughout the year, one for each financial quarter and one annual meeting. These meetings are where all major decisions are made about the company and the direction it is heading. We propose Sound Credit Union to put the investment decision on adopting Hyper-Personalization practices to vote for its members. We will explain to members how this technology adoption will be beneficial to both the bank and its customers. We will engage with members regularly on the impact of the changes during the subsequent board meetings.

Key outcomes: Board members are brought in on the proposal plan and budget

Member Feedback Loop

APR 2023

We will begin our solution strategy with a small targeted pilot on a fundamental function for SCU. We will only enlist a small sized opt-in users who are willing to participate in the pilot program. Through the pilot, we want to prove the concept to Sr Leaders of the company on the effectiveness of hyper-personalization with no compromises to the robust privacy and security of users. We will be transparent in our research purpose with the cohort of users by providing a full view of what data of the members were used, and how. The pilot should also provide a validation of the efficacy of the solution, and learnings that we will be able to apply towards the full solution.

ey outcomes: Pilot hyper-personalization, and demonstrate value

Community Events:

MAR 2023

It was resoundingly evident from the surveys that we need to better carry the message on SCU's strong privacy practices to their customers to build better confidence and trust. We will host a series of community outreach events and social media campaigns to reaffirm customers' trust on SCU data collection practices. This will help build trust in members by showing a commitment to the community, enhancing communication, building relationships with customers, helping customers understand and manage their finances, and by being accessible.

Key outcomes: Develop a sound communication plan for community outreach

Optimize

Data Collection

Objectives: Expand data collection methods

In this phase of the project, we focus on building the right foundations towards the implementation of the personalization models. ML requires high quality data that is essential to the success of the model's precision and quality. We will also focus on a sustainable model to managing operations for SCU.

Optimize the Data Ingestion Models

When activities will start

MAY 2023

Application of ML is largely dependent on sound data quality practices to train the models. Poor data quality results in inaccurate predictions and poor models. We will inventorize current processes for data collection at various transaction events - both online and offline. The objective is to make the data ingestion models canonical, consistent, and real-time to apply ML personalization. We will apply anonymization of data where needed to ensure the highest levels of customer privacy and apply AI techniques to understand customer needs which will result in better personalized recommendations for SCU's customers.

Key outcomes: Address data quality of the current pipelines to prepare for ML

Training Existing Staff:

JUN 2023

From the outset, we aimed to create a sustainable solution. Hiring data analysts can significantly help Sound optimize their data collection methods by having a dedicated team to extract insights from the data collected, which can inform business decisions and strategies, however the cost to hire these employees too high for our planned budget. Instead, we plan for Sound to upskill 4 existing business analysts on ML solutions. This new data analysis team can also utilize their data visualization and statistical modeling skills to identify trends/patterns in customer behavior and market trends for improved data collection efforts.

Key outcomes: Create a sustained model for operations at SCU

Rollout of the Revamped Personalization Model:

JUL 2023

Machine learning will allow Sound Credit Union to automate its hyper-personalization as we move towards a digital-focused company structure. We are planning to implement a specialized form of machine learning called augmented analytics. This means that it allows for real-time data analysis, ultimately allowing us to identify key drivers of certain changes. We plan to outsource this ML software to reduce the budget of the proposed plan.

Key outcomes: ML models are created towards hyper-personalization

Network

With Underserved Communities

Objectives: Use hyper-personalization for better customer targeting

In this phase, we will build reach out measures towards new market segments for underserved and student communities to bring SCU closer to their mission.

When activities will start

Help Students:

FEB 2023

Students are significantly underserved in the banking industry, due to lack of financial education. Over half of college students in the United States cannot answer basic financial literacy questions. To assist this demographic, we plan to offer financial education services on budgeting and debt management, promoting community welfare and strengthening connections with potential Sound customers. With this support, we hope to foster future banking relationships with Sound. Sound can also offer low-fee student accounts, user-friendly online tools, and career development events at universities to aid them.

Key outcomes: Create engagement with community on financial literacy /support

Support Service Workers:

MAR 2023

We plan to assist all types of service workers, including teachers, police officers, doctors, veterans, and others, as they are often looked over, not given as many opportunities as others, and are susceptible to debt and predatory practices, which can result in financial mismanagement and severe consequences for their lives. Sound can make a significant contribution to the community by assisting these workers in promoting long-term stability, with services tailored to their needs. Supporting them is an excellent way for Sound to give back to the community. We will offer flexible payment options and collaborate with community organizations to provide financial assistance to them.

Key outcomes: Prepare a positive outreach and engagement

Target Low-Income Demographics:

JUL 2023

The third target community is low-income individuals, who are unfairly disadvantaged in managing their finances. Most financial institutions are unwilling to conduct business with these people and even if they do, those in the community face exorbitant transaction fees. As the most vulnerable community in the financial industry, Sound can follow their goal to help the community by targeting these individuals. Sound can use hyper-personalization services to target these demographics through customer segmentation and targeting technologies. Sound can help these communities by offering low-cost or no-cost banking with reduced interest rates, financial education and counseling, and credit-building programs.

Key outcomes: Tailored solutions for underserved communities

Drive

For Scale with Hyper-personalization

Objectives: Develop AI/ML methods to apply hyper-personalization

In this phase, we unleash the capabilities of hyper-personalization to the overall banking operations and building a cohesive experience on the customer facing portals of SCU

When activities will start

Deliver Hyper-Personalization at Scale

MAR 2023

We will apply the successes of the pilot in delivering hyper-personalization at scale during this phase. The intent is to make hyper-personalization through AI services as a fabric through which the various pipelines - namely marketing, websites, and customer service contact points, are delivered bringing in a tighter integration to apply hyper-personalization. We will roll out the revamped services and experiences on campaigns on a wider user base for greater gains for SCU. We can also use machine learning algorithms for fraud detection, risk management, and Anti-Money Laundering (AML) software.

Key outcomes: Rollout full suite of hyperpersonalization tools

Streamline Operational Processes

MAY 2023

We propose to unleash our integrated 360° approach to the key processes in SCU's operations here. We will build our updated personalization engine into other complementary areas of the bank where permissible. One example is to integrate the data pipelines with the existing CRM solutions to drive better tailoring for prospect management. Another key application is to revamp the social media integration channels to provide greater efficiency by bringing the benefits of ML-based models. With real-time data collection and integrated processes, we believe this will bring a step change improvements in the bank's outreach and marketing material.

Key outcomes: Provide additive value of ML enabled operations

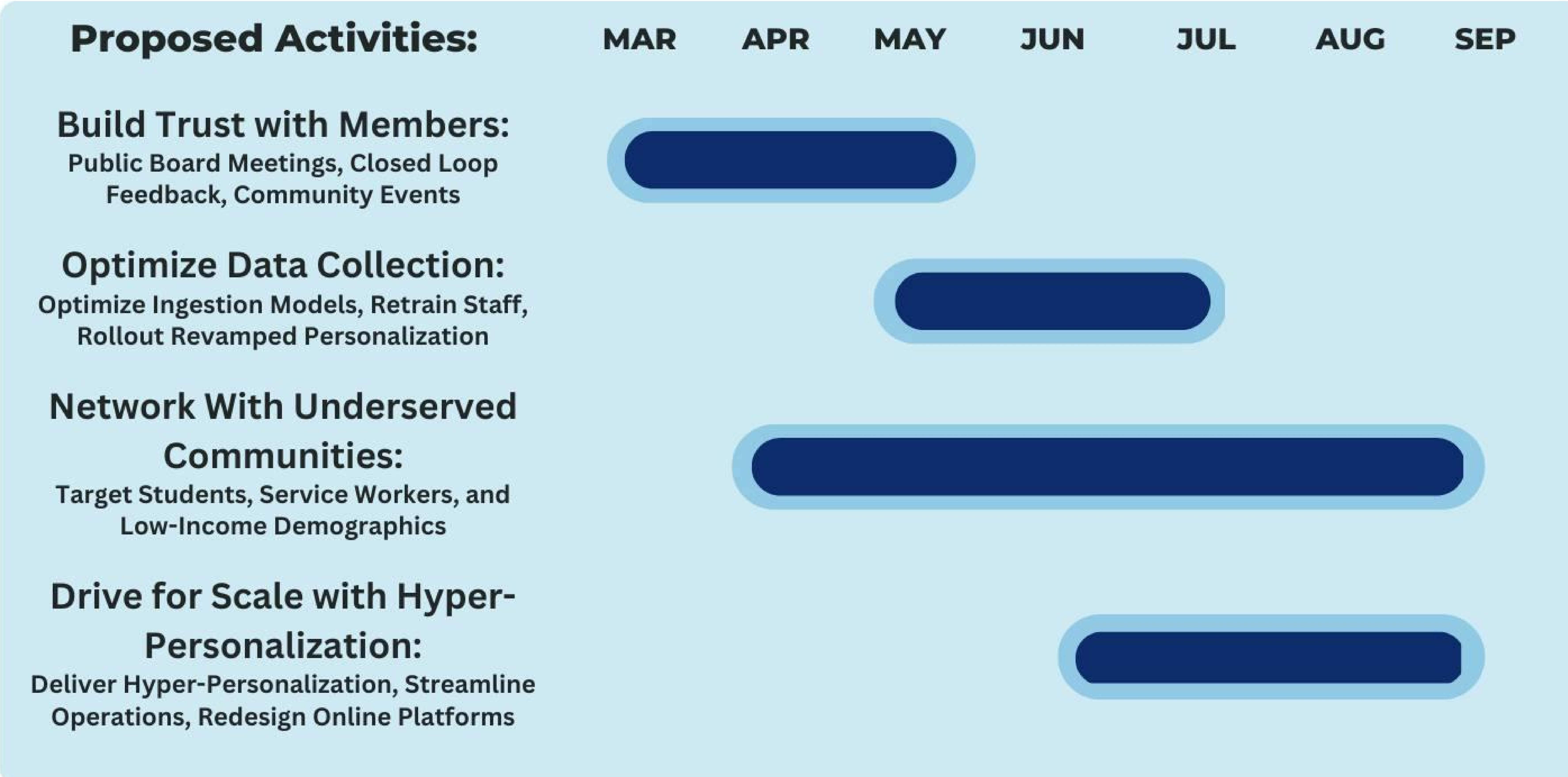
Redesign Sound's Online Platforms:

JUN 2023

To deliver a richer experience for members, we will revamp Sound's customer facing experiences. The website and mobile application are the primary entry points for customers, and we plan to enrich this via useful prompts to make the customer experience better (eg. recurring charges, did you want to stop the subscription?), chat-GPT style "ask me anything" feature providing real-time answers/suggestions to answer customer inquiries. We will also redesign Sound's social media presence to better attract potential customers and specifically target proposed demographics through channels like Instagram and Facebook. This phase delivers the culmination of Sound's hyper-personalization initiative.

Key outcomes: Synergize platforms at large towards hyper personalization

2023 Proposed Activities Timeline:



C. Proposed metrics and key performance indicators to measure plan effectiveness

General KPIs

Metric	Description
Customer Engagement • Click-through rates and follow-through rates on customer recommendations • Net new services added for existing customers • Abandonment rate	Tracks level of engagement to the proposed data personalization program. We intend to measure signups, and general uptake of the recommendations provided through hyper-personalization.
Customer Feedback • CSAT on data collection policies through surveys • NPS overall on bank's performance	Collecting feedback from customers through surveys, focus groups or other means to understand their satisfaction with the personalization experience and identify areas for improvement.
Customer Acquisition/Retention Metrics • Net new customers added	Tracking new customer signups/retention per target demographic. Higher numbers indicate attribution due to the plan.

To measure the impact of the BOND initiative and track our progress over time, we will be monitoring several key performance indicators (KPIs). These targeted metrics are specifically designed to evaluate the performance of our organization and the activities of the BOND initiative. We will use these KPIs to assess the level of interest in the credit union, measure the effectiveness of our customer feedback and targeting strategies, and track website traffic, mobile app downloads, and social media engagement. By measuring these primary performance indicators for each stage of the proposed plan, we will be able to accurately gauge the success of the BOND initiative. We will track the metrics over defined period intervals (WoW, MoM, YoY), and use them to fine-tune/adjust algorithms used in the program.

	Activities	Performanc e Indicator(s)	Rationale	Calculation	Benchmarks
B	• Public Board Meetings • Community Outreach Events • Opt-in rollout for a small cohort	• Member Participation Rate • CSAT • NPS among cohort users	This KPI can be used to gauge the level of interest and engagement in the bank's activities	[(# of Participating Members) / (# of Total Members)] X 100	Underperforming: 5-10% On Target: 10-15% Exceeding Expectations: >15%
O	• Customer Surveys	• Click-Through Rate • Customer Engagement Rate	This KPI can be used to see how effective the surveys are at determining customer feedback	[(# of Responding Members) / (# of Total Members)] X 100	Underperforming: 5-10% On Target: 10-15% Exceeding Expectations: >15%
N	• Targeting students, low-income individuals, service workers	• Net new signups / demographic	This KPI can be used to see how effectively Sound can target specific demographics	[(# of Underserved Demographics Post-Plan) / (# of Underserved Demographics Pre-Plan)] X 100	Underperforming: 5-10 % On Target: 10-15% Exceeding Expectations: >15%
D	• Redesigning website, app, and social media presence	• Reviews/ Engagements collected on website (Google), and App Stores (Apple, Android)	This KPI can be used to measure website traffic, mobile app downloads, and social media engagement	NPS > 7 (currently 5) App reviews > 4 stars (currently 3)	Underperforming: <= 7 (NPS), < 4 stars (App) On Target: 7 (NPS), > 4 stars (App) Exceeding Expectations: > 9 (NPS), > 4.5 stars (App)

VI. PROPOSED BUDGET

Hyper-personalization brings transformational upgrades to the banking operations, and directly yields in increased revenue, and customer satisfaction. We propose a phased model towards hyper-personalization that allows SCU to iteratively apply learnings towards building the product. Our approach combines (1) foundational revamp and education of the bank's direction to build trust and (2) implementing technology solutions driving personalization. Perry Maxfield, the VP of Digital and Innovation Delivery at Sound, gave us a budget between $250-275K to implement our proposed plan.

The cost structure for the proposed solution is shown below:

	Budgetary Estimated Activities:Specific Costs:			Elements:Costs for 2023:
B	• Community Events • Marketing Campaigns • Hyper-personalization pilot	• Venue • Marketing • Social Media Campaigns	• $400 / event * 3 events • $20K for ad and mail campaigns	$21,200
O	AI integrated data collection, and analysis	• Marketing campaigns (social channels and recommendations	• Software Licensing: $200/month • AWS Costs: $10k/month	$124,200
N	• Targeting students, low-income individuals, service workers	• Tailor financial services through AI • Free Financial Workshops	• $15 per staff per hour	$18,000
D	• Redesign platforms with AI integration	• Using AI recommendations for website / app experience	• Staff costs: $100K	$100,000

BOND Total Cost for 2023: $263,400

Based on our research, we estimate the size of opportunity with expanding customer segments leading to new customers in the range of 250k - 1.5M customers, and the downstream impact of hyper-personalization resulting in > 15% in incremental revenues for the bank. In addition, employment of AI technologies will result in reduction of material threats, and lowering compliance costs through technology solutions by 15-20%. Summarily, the revamped solution would lead towards a more competitive solution, increasing revenues, and lowering costs, with a projected ROI of XXX%. We compute the conservative targets on ROI as follows:

Return on Investment

Customer Acquisition

We expect our hyper-personalization solution to drive better outreach towards customers through better targeting and personalization of programs. Our transparent messaging, and community-education on the privacy and compliance should drive appeal for new customers. Our solutions through networking with student, and underserved communities driving , and specialized banking programs should drive greater enrolment.

Increased Spending per Member

One of the direct influences of our hyper-personalization solution is to drive specifically catered programs that drive greater participation among existing members. We will introduce specific programs that appeal to demographics and segmentation models in the areas that SCU operates in - an example is introducing money transfer solutions given the high concentration of an expat community among SCU's user base. We remain confident that our personalized banking services, and cross-sell abilities driven through hyper-personalization will enable propensity towards upsell/cross-sell services.

$$\frac{\text{Net Profit (Revenue - Expenses)}}{\text{Investment (Expenses)}} = \frac{\$500{,}000 - \$263{,}400}{\$263{,}400} = 0.898 =$$

90% ROI

VII. BIBLIOGRAPHY

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